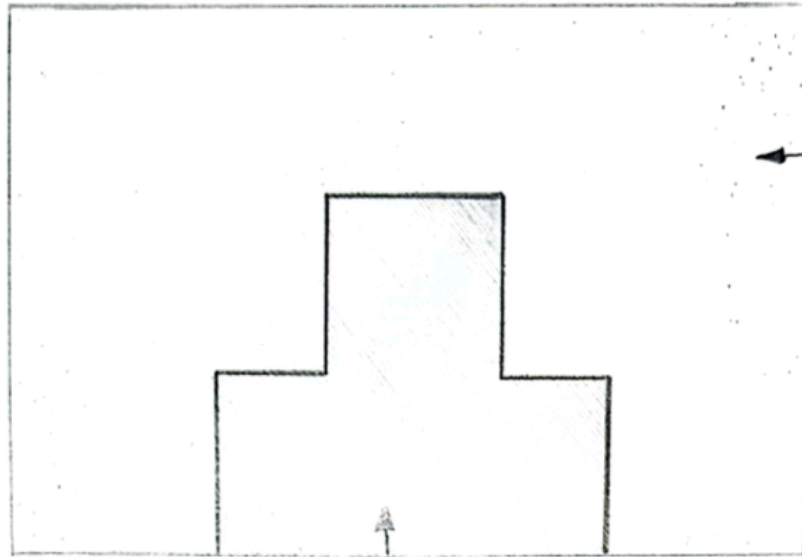
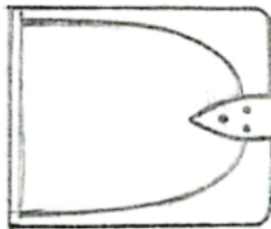


FOUNDRY
SHOP

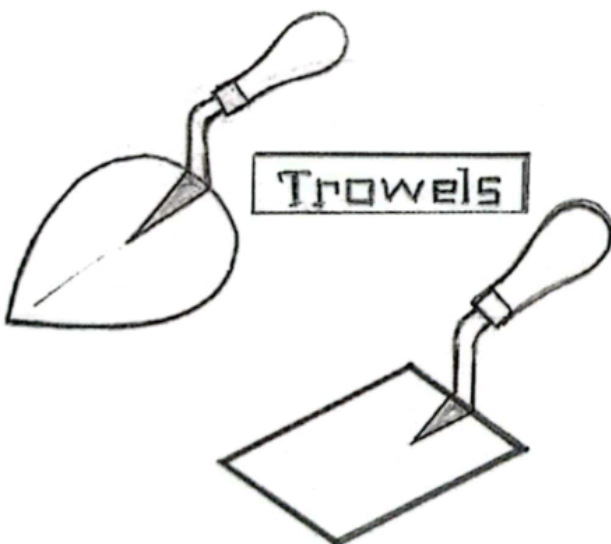
FRONT VIEW of JOB PATTERN



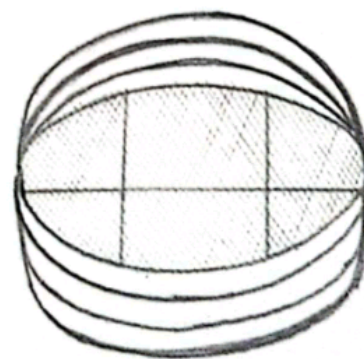
Pattern



Shovel



Trowels



Hand Riddle



HARCOURT BUTLER TECHNICAL UNIVERSITY

KANPUR

WORKSHOP PRACTICE

FULL NAME Priyanshu CLASS B.S.MS [1st Year]

SHOP Foundry Shop DATE

PRACTICE-1st

Objective: To prepare a simple mould of the given pattern in green sand.

Date of Starting: 03.02.2026

Date of Finishing: 03.02.2026

★ Materials Required: Pattern, Green Sand, Parting Sand, Water.

★ Tools and Equipments Used: • Shovel • Hand Riddle • Trowels
• Rammers • Smoother • Spirit Level • Moulding Box • Sprue Pins
• Side pins • Vent wire • Lifter and Cleaner • Draw Spike.

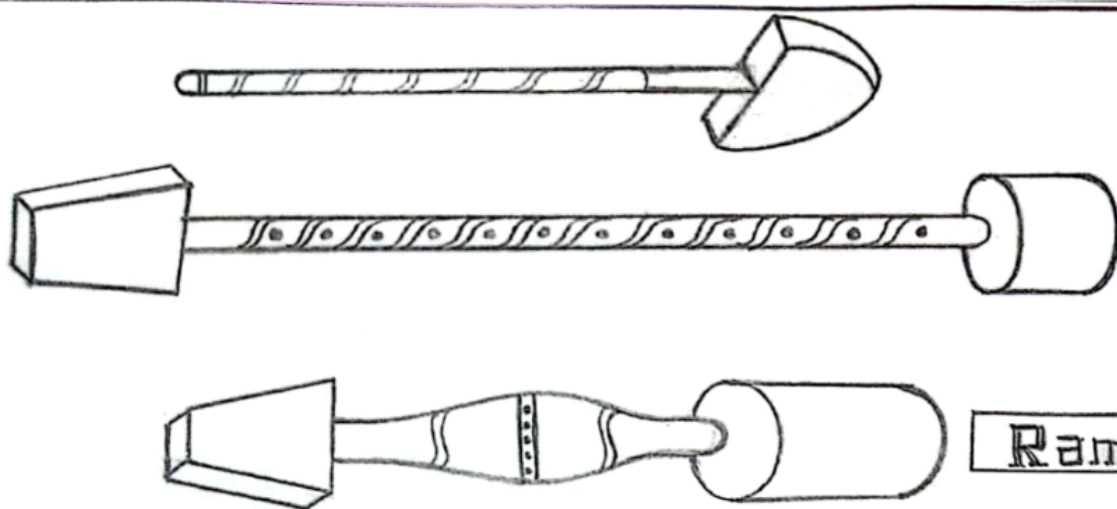
★ Description of Tools and Equipments Used ★

* **Shovel**: It consists of an iron pad fitted with a wooden handle. It is used in mixing and conditioning the foundry sand by hand and transferring it into the flask.

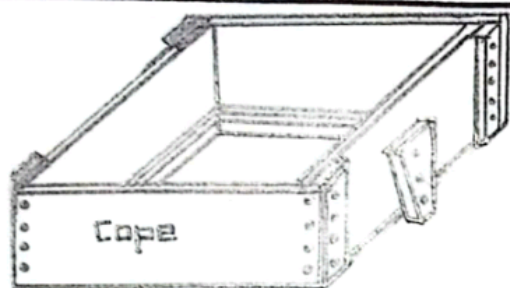
* **Trowels**: Trowels are used for finishing flat surfaces and joints in a mould. The common shapes of are shown. They are made of iron and are provided with a wooden handle.

* **Hand Riddle**: It consists of a wooden frame fitted with a screen of standard wire mesh at its bottom.

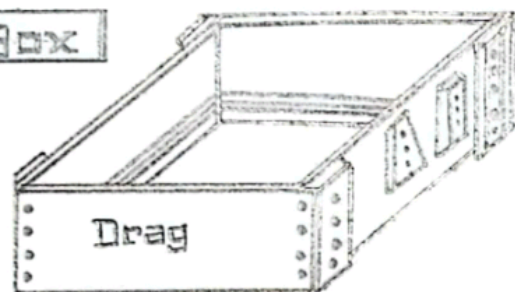
Paras



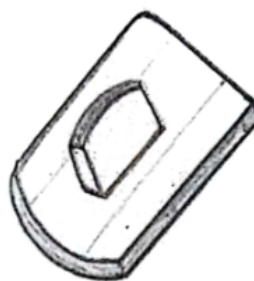
Rammers



Moulding Box



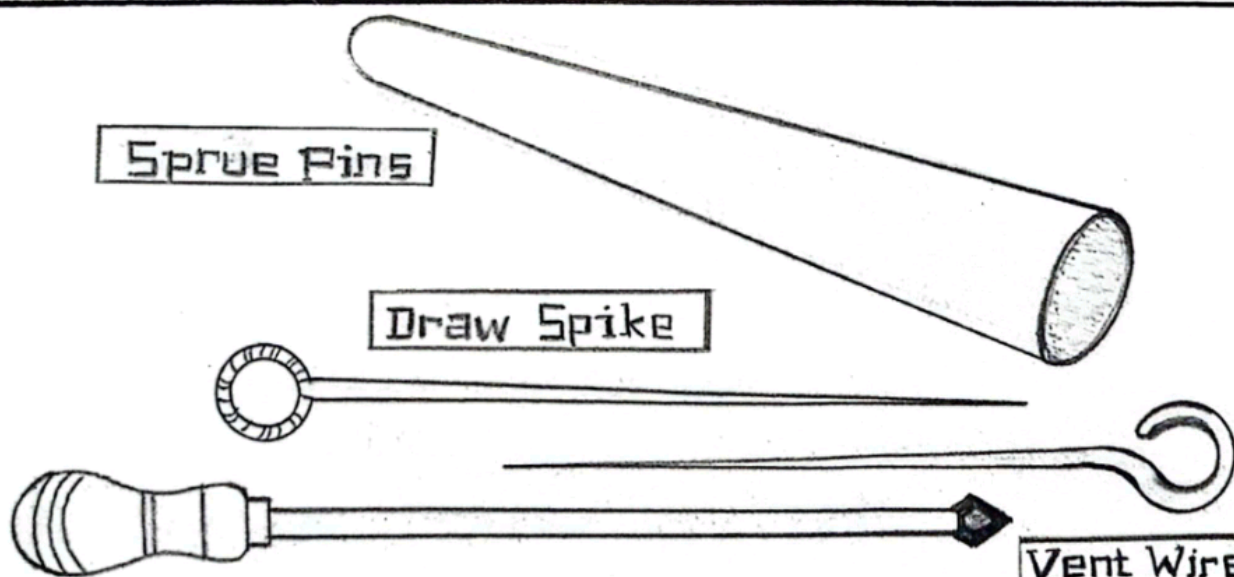
Drag



Smoother

Sprue Pins

Draw Spike



Vent Wire

* **Rammers:** Rammers are used for striking sand mass in the moulding box to pack it uniformly around the pattern.

The common forms of rammers used in hand ramming are following:

- **Peen Rammer:** It has a wedge shaped construction formed at the bottom of a metallic rod.
- **Hand Rammer:** It is smaller than peen rammer, on one end it carries a wedge type construction called peen and on the other a solid cylindrical shape, klas butt.
- **Floor Rammer:** It carries a peen at one end and flat portion on other end.

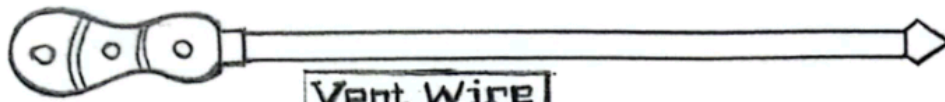
* **Smoother and Corner Flicks:** They are finishing tools used for repairing and finishing flat and round surfaces round or square corners and edge.

☼ **Moulding Boxes or Flasks:** The moulding boxes used in sand are 2 type

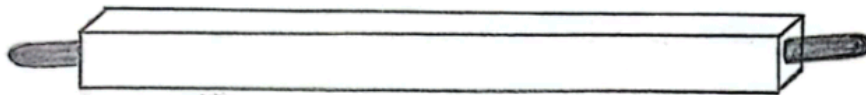
- **Closed Moulding Boxes:** They are a kind of closed ones - modern.
- **Open type or Snap Flasks:** It is made of wood and hinged at one corner. These boxes consist of two or more parts. The lower part is called **drag**, the upper part **cope** and all intermediate parts if used called **cheeks**.

* **Sprue Pins:** It is a tapered rod of wooden or iron which is embedded in the sand and later withdrawn to produce a hole called **runner**, through which the molten metal is poured into the mould.

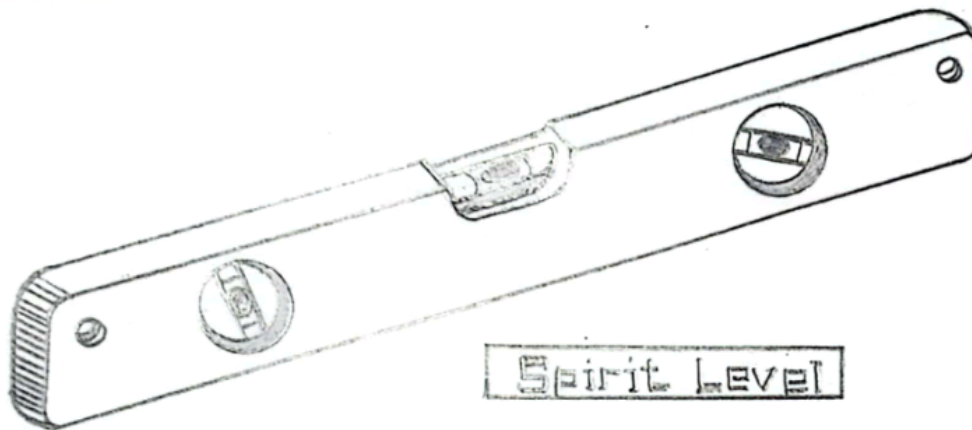
* **Draw Spike:** It is a tapered steel rod having a loop or ring at its end (one end) and a sharp point at the other. It is used to rap and draw pattern from the mould.



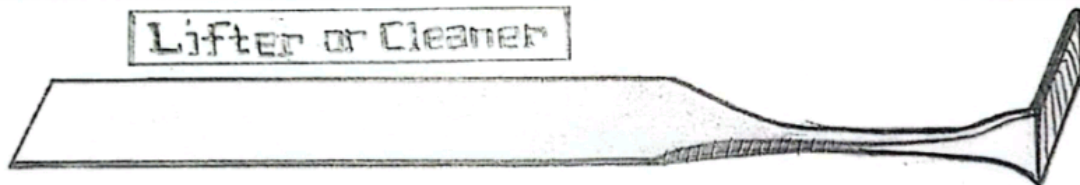
Vent Wire



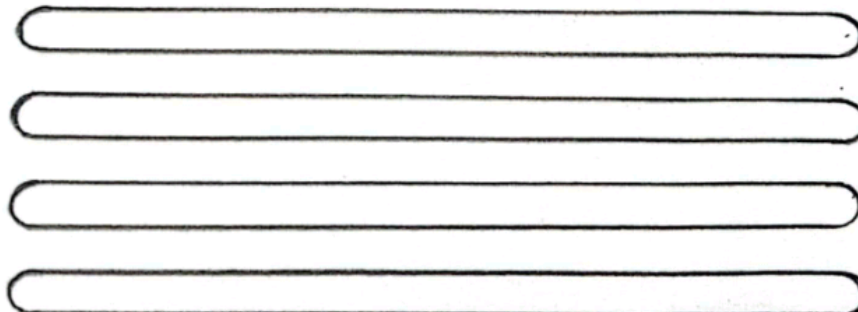
Strike-off Bar



Spirit Level



Lifter or Cleaner



Side Pins

Priyanshu
250137040
B.S.-M.S.
Mathematics
and
Data Science

* **Vent Wire**: It is a thin steel rod or wire carrying a pointed edge at one end and a wooden handle or bent shape loop on the other. After ramming and striking off the excess sand it is used to make small holes called vents in the sand mould to allow the exit of air or gases and steam during casting.

* **Strike-off bar**: It is a flat bar made of wood or iron to strike off the excess sand from the top of a box after ramming.

* **Spirit Level**: It is used to check if a surface is perfectly horizontal (level) or vertical (plumb) by centering a bubble in a vial of liquid.

* **Lifter and Cleaner**: They are finishing tools are used for repairing and finishing the sand mould after withdrawal of pattern. They are also used for removing loose sand from the mould cavity. The useful forms are:

- Simple Lifter and Cleaner
- Yankee lifter.

* **Side Pins**: It is thick pin made of metal which is used to locate the position of moulding box, so that the moulding box do not get misplaced.

PATTERN

★ A pattern may be defined as replica model of desired casting which when packed or embedded in a suitable moulding material produces a cavity called mould.

❖ The way in which a pattern differs from actual components are:

* It carries an additional allowance to compensate for metal shrinkage.

* It carries the necessary draft to enable its easy removal from sand mass.

★ **Pattern Material:** It is the common material of which the pattern are made, these are following:

★ **WOOD:** It is the most common material used for pattern making because of following advantages:—

- It is cheap and abundantly available.
- Its manipulation is easier.
- It gives good surface finish.
- It can be easily shaped.

* Wood has certain disadvantages are:

- Wears out quickly.
- It is very susceptible to moisture
- Its life is shorter.

The common woods are used :

• Pine • Teak • Shisham • Mahogany • Kail • Deodar

METALS: Metals are also used when a higher dimensional accuracy is needed. They have a longer life than wooden ones.

Metals have some disadvantages:

They are costlier, and Heavier.

Need machining for surface to be fine.

They have tendency to get rusted.

The common metals are used as metallic pattern are:

• Cast Iron • Brass • White metal • Aluminium and its alloys.

PLASTER OF PARIS: It is used because it can be easily casted into intricate shapes.

Controllable expansion.

Highly compressive strength.

Expands on being solidified.

PLASTIC: Plastics are gradually gaining favour as pattern materials due to their following specific characteristics:—

Lighter in weight.

High resistance to wear.

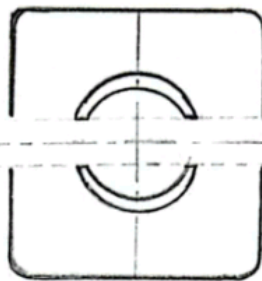
Fine surface finish.

High strength.

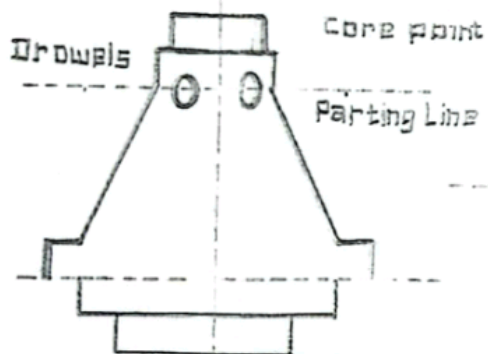
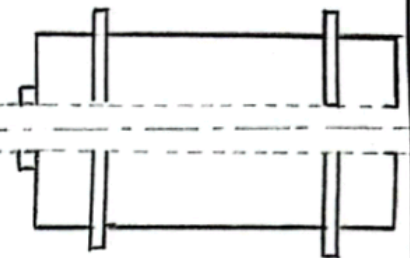
WAX: Wax patterns are exclusively used in investment casting. for this a die or metal mould is made in two halves into which the heated wax is poured.



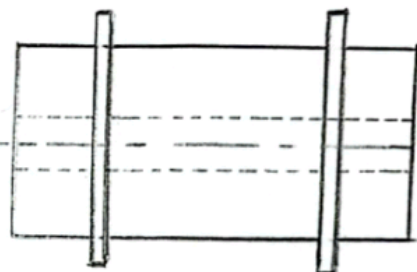
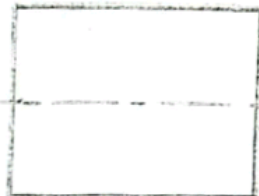
Solid Pattern



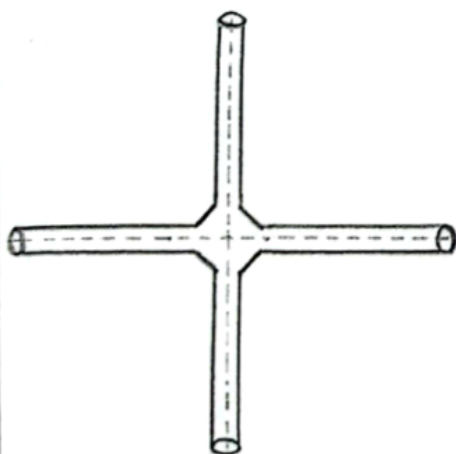
Two Halves of Pattern



Three Part Pattern

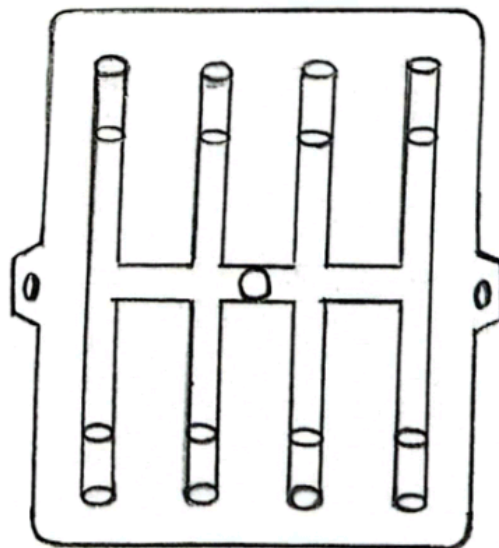


Prepared Casting



Gated Pattern

Match Plate Pattern



★ **Types of Pattern:** The type of pattern to be used for particular casting depends upon many factors like the bulk of casting, type of moulding process, number of castings, required and anticipated difficulty of moulding on account of the typical shape.

- i) Solid or single piece pattern ii) Two piece or split pattern
- iii) Multipiece pattern iv) Match Plate pattern v) Gated Pattern
- vi) Skeleton pattern vii) Sweep pattern viii) Pattern with loose pieces
- ix) Cope and drag pattern x) Follow board pattern
- xi) Shell pattern xii) Segmental pattern.

★ **Solid or Single Piece Pattern.**

- Simplest of all patterns. • One piece.
- Cheaper pattern. • Manually operated.

★ **Two piece or split pattern**

- Have two parts. • One part in drag and other by cope.
- Complex than previous one.

★ **Multipiece pattern**

- Used in complicated designs. • Contains 3-4 parts or even more.
- It is more complex.

★ **Match plate pattern**

- It is accurate. • Mainly used in Machine moulding.
- Expensive construction cost.

★ **Gated Pattern**

- Used in mass production • Multicavity moulds needed.
- Gates formers are used.

★ **Skeleton Pattern:** • Consists of wooden frames and Strips

- Filled with loam sand and rammed • Surplus sand is removed by strickle.

Pattern Allowances

A pattern may be larger than required size of the casting in order to allow for various factors, such as shrinkage, machining, distortion, and rapping etc. The following allowance are provided in a pattern are given below;

1. Shrinkage Allowance:

Most of the metals used in casting work contract during cooling from pouring temperature to the room temperature, in three form

• Liquid Contraction • Solidifying Contraction • Solid Contraction.

The first two are compensated by gates and risers and the last one by providing adequate allowances in the pattern.

The amount of contraction varies with different metals and therefore, their corresponding allowances also differ.

2. Machining Allowance:

A casting may require machining all over or on certain specified portions, depending upon the assembly conditions and the operation it has to perform. Such portions or surfaces are marked duly in the working drawings.

3. Draft Allowance:

All patterns are given a slight taper on all vertical surfaces, i.e., the surfaces parallel to the direction of their withdrawal from the mould. This taper is known as draft or taper allowance.

4. Rapping or Shake allowance:

When a pattern is to be withdrawn from the mould, it is first rapped or shaken, by striking over it from side-to-side, so that the surface may be free of adjoining of sand wall of the mould. As a result of this the size of mould cavity increases a little and a negative allowance is to be provided in the pattern to compensate the same.

5. Distortion Allowance:

The tendency of distortion is not common in all the castings. Only casting which have an irregular shape and some such design that the contraction is not uniform throughout will distort during cooling on account of the setting up the thermal stresses in them.

★ PROCEDURE ★

Prepare the moulding sand for moulding process and make sure it is not much moist

Make a pit in the sand to place the pattern.

Rub the sand in between the palms until it becomes finely grained.

Fill the sand in the pit and make sure that pattern must lie in the upper center.

Use the rammer according to the pit size and check the level by spirit level.

Page No. _____

Date _____

Cover the surface of pit with parting sand

Place the cope box over the drag i.e., pit.

Make a small layer of facing sand in the cope box over parting sand

Fill the cope box with sand and properly use the rammer over it.

Place the side pins in opposite corner of cope box in outer side.

Do the ramming and finishing of cope box by placing the sprue pin near the pattern with the help of smoother.

Take out the sprue pins properly.

Make the funnel in sand with the help of lifter, separate the cope and the drag.

Make the gating system with the help of lifter.

The pattern is taken out with the help of draw spike, moulding is completed.

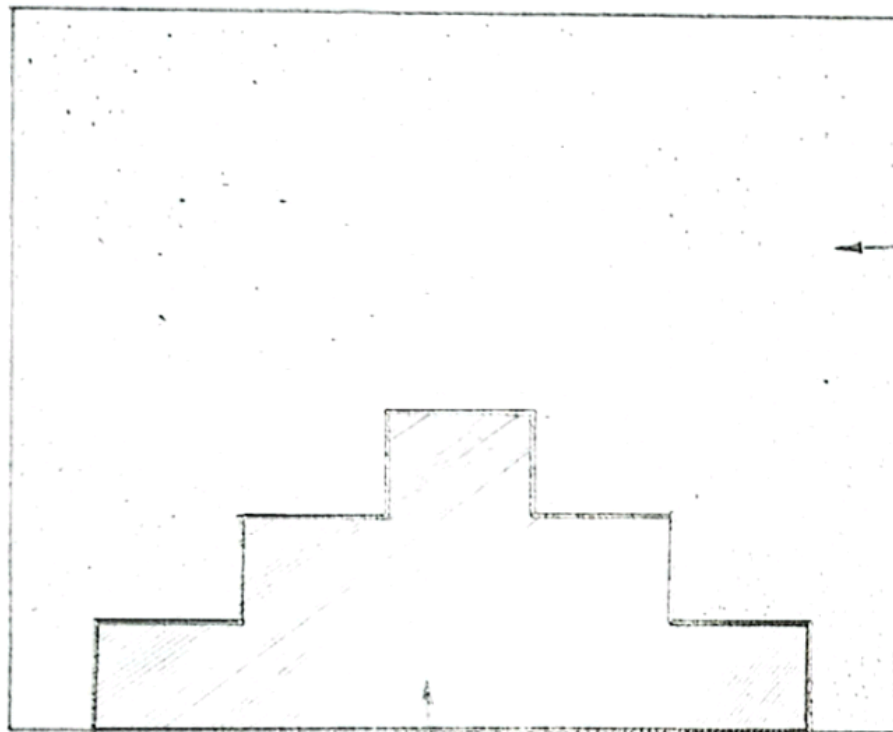
★ PRECAUTIONS ★

Maintain the moisture in the sand.

Use the rammer with constant force.

Carefully separate out cope box with pit.

Carefully take out the pattern from pit.



Green
Sand

Pattern

FRONT VIEW OF TWO STEP JOB PATTERN

Priyanshu
280137040
B.S.-M.S.
Mathematics
and
Data Science



HARCOURT BUTLER TECHNICAL UNIVERSITY

KANPUR

WORKSHOP PRACTICE

FULL NAME.....Priyanshu..... CLASS.....B5-M5. [1st Year].....

SHOP.....Foundry Shop..... DATE.....

PRACTICE-2nd

- ★ **Objective:** To prepare a mould with two step pully in green sand with runner and riser.

Date of Starting: 10-02-2026

Date of Finishing: 10-2-2026

★ Material Required★

- Pattern
- Green Sand
- Parting sand
- Water.

★ Tools and Equipments Used★

- Shovel
- Hand riddle
- Trowels
- Rammers
- Smoother
- Spirit level
- Moulding box
- Vent wire
- Sprue pins
- Side ping
- Lifter & Cleaner
- Draw spike

★ Moulding Sand★

The common source of collecting foundry sands are rivers, lakes, sea, deserts.

* Types of Moulding Sand *

- Green Sand
- Dry Sand
- Parting Sand
- Facing Sand
- Floor or baking Sand
- Core Sand
- Oil Sand
- Molasses Sand.

Page No. _____

Date _____

★ Green Sand ★

It is also known as tempered sand. It denotes a well prepared foundry sand which contains just enough moisture to give it sufficient bond. Moulds in this sand are known as green sand moulds and do not required any baking before pouring the molten metal in to them.

* Composition of Green Sand:

- 80 % - 90 % .-- Silica sand (SiO_2)
- 10 % - 20 % .-- Clay
- 2 % - 8 % .-- Water

★ CHARACTERISTICS OF FOUNDRY SANDS ★

★ **Refractoriness:** It is that property of the moulding sand which enable to resist high temperature of the molten metal without fusing, thus facilitating a clean casting.

★ **Permeability:** It is also termed a porosity. It is that property of sand which allows the gases and steam to escape through the sand mould. If these gases are not allowed to go out they will either make the casting unsound or blast the mould.

★ **Flowability or Plasticity:** It is that property of the sand that due to which it flows during ramming to all the portions of the moulding flask, packs properly around the pattern to acquire the desired shape and distributes the ramming pressure evenly to all parts of the mould.

Adhesiveness: It is property of the sand due to which it is capable of adhering to the surfaces of other materials. It is entirely due to this property that the heavy sand mass is successfully held in a moulding flask and manipulated as desired without any danger of its falling down.

Cohesiveness: It is that property of the sand due to which its rammed particles bind together firmly so that the pattern is withdrawn from the mould without damaging mould surfaces & edges.

Collapseability: It is the property due to which the sand mould automatically collapses after solidification of the casting to allow a free contraction of the metal.

MOULDS: A mould can be defined as a void created in a compact sand mass which when filled with molten metal, will produce a casting. The process of producing this cavity or mould is called moulding.

Types of Moulds *

- | | |
|----------------------|---------------------|
| ● Green Sand Moulds. | ● Dry sand moulds. |
| ● Skin dried moulds | ● Air dried moulds. |
| ● Loam moulds | ● Plaster moulds. |
| ● Core sand moulds | ● Metallic moulds. |
| ● Shell moulds. | |

Page No. _____

Date _____

★ MOULDING ★

Moulding is process of manufacturing by shapeing liquid or pliable raw material using a rigid frame called mould. This itself may have been made using pattern or mould of the final object.

Types of Moulding.

- Floor moulding ● Bench moulding ● Pit moulding ● Shell moulding
- Machine moulding ● Plaster moulding ● Metallic moulding.

★ CORE ★

Core can be difined as a body of sand generally prepared in a mould box which is used to form a cavity of desired shape.

Types of Core

- Horizontal core ● Vertical Core ● Balance Core
- Hanging or cover core ● wire core.

★ PROCEDURE ★

- * Prepare the moulding sand for moulding process and make sure it is not much moist.
- * Make a pit in the sand to place the pattern.
- * Rub the sand in between palms until it become finely grained.
- * Fill the sand in the pit and make sure that the pattern must be lie in the upper center.
- * Use the rammer according to the pit size.
- * Check the level by spirit level.
- * Cover the surface of pit with parting sand and place the cope box over the drag i.e., pit.
- * Make a smaller layer of facing sand in the cope box over parting sand.

- * Fill the cope box with the sand and properly use the rammer over it.
- * Place the side pins in the opposite corner of cope box in outer side.
- * Do the ramming and finishing of cope by placing the sprue pins near the pattern with the help of smoother.
- * Take out the sprue pin properly.
- * Make the funnel in the sand with the help of lifter and cleaner.
- * Separate the cope and the drag.
- * Make the gating system with the help of lifter.
- * The pattern is taken out with the help of draw spike.
- * Moulding is completed.

★ PRECAUTIONS ★

- * Maintain the moisture in the sand.
- * Use rammer with constant force.
- * Carefully separate cope box with pit.
- * Carefully take out the pattern from pit.